

## **1. Administrative & Study Identification**

**Full Study Title:** A Global, Multicenter, Randomized, Double-blind, Double-dummy, Parallel-group, Phase 3b Study to Assess the Efficacy, Safety, and Tolerability of Remibrutinib 25 mg b.i.d. in Comparison to Placebo With Omalizumab 300 mg Every 4 Weeks as Active Control Over 52 Weeks in Adult Patients With Chronic Spontaneous Urticaria Inadequately Controlled by Second Generation H1-antihistamines and an Open-label 52-week Optional Extension to Assess Long-term Efficacy, Safety and Tolerability of Remibrutinib 25 mg b.i.d. **Short Title/Acronym:** Phase 3b Study to Assess the Efficacy, Safety, and Tolerability of Remibrutinib in Comparison to Placebo, With Omalizumab as Active Control, in Adult CSU Patients, Followed by an Open-label 52-week Optional Extension. **Protocol Number:** CLOU064A2304 **NCT Number:** NCT06042478 **Posting ID:** 139866335997781069 **Trial Status:** Active, not recruiting (ACTIVE\_NOT\_RECRUITING) **Sponsor & Company Name:** Novartis / Novartis Pharmaceuticals / Novartis Pharma AG **Investigational Product:** Remibrutinib (LOU064) **Phase of Development:** Phase IIIb (PHASE3) **Medical Monitor:** Novartis Clinical Operations Lead / Contact: Barbara Roemer

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## **2. Study Synopsis & Rationale**

**2.1 Study Objective Primary Objective:** To demonstrate that remibrutinib (25 mg b.i.d.) is superior to placebo in adult Chronic Spontaneous Urticaria (CSU) participants with respect to the absolute change from baseline in the Weekly Urticaria Activity Score (UAS7), Weekly Itch Severity Score (ISS7), and Weekly Hives Severity Score (HSS7) at Week 12. **Secondary Objectives:** To evaluate the long-term efficacy, safety, and tolerability of remibrutinib up to 52 weeks compared to both placebo and the active control omalizumab (300 mg q4w).

**2.2 Background & Rationale** To address the debilitating symptoms of CSU—wheals and hives—in patients whose condition is inadequately controlled by second-generation H1-antihistamines (H1-AH). Remibrutinib is designed to reduce the severity of itch and hives by inhibiting a protein that triggers the cellular release of histamine, proteases, and cytokines. This study seeks to directly compare remibrutinib to the standard-of-care biologic omalizumab and placebo to establish its clinical utility, speed of onset, and sustained efficacy.

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### 3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Placebo-controlled and Active Comparator (Omalizumab) **Blinding:** Double-blind, Double-dummy (using matching placebo injections and tablets) **Randomization:** Randomized, Parallel-group

**3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$):** 471 participants (approximately 468 planned for randomization) **Number of Sites:** Multicenter (Across 24 countries globally, including EU/EEA locations) **Estimated Study Start:** 15-Feb-2024 **Estimated Primary Completion:** August 2025 (Estimated global end of trial: 08-Jul-2027)

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### 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Male and female adult participants  $\geq 18$  years of age at the time of signing the informed consent. 2. Diagnosis of Chronic Spontaneous Urticaria (CSU) inadequately controlled by second-generation H1-antihistamines (H1-AH) at the time of randomization. 3. CSU duration for  $\geq 6$  months prior to screening, with the presence of itch and hives for  $\geq 6$  consecutive weeks prior to screening, despite the use of second-generation H1-AH during this time period. 4. UAS7 score  $\geq 16$ , ISS7 score  $\geq 6$  and HSS7 score  $\geq 6$  during the 7 days prior to randomization (Day 1), and documentation of hives within three months before randomization. 5. Willing and able to complete an Urticaria Patient Daily Diary (UPDD) for the duration of the study and adhere to the study protocol. Participants must not have had more than one missing UPDD entry (either morning or evening) in the 7 days prior to randomization (Day 1).

**4.2 Exclusion Criteria** 1. Prior exposure to ligelizumab, omalizumab, and other biologics with any effect in CSU, including anti-IgE therapies. 2. Significant bleeding risk or coagulation disorders, requirement for anti-platelet or anti-coagulant medication, history of gastrointestinal bleeding, or history/current hepatic disease. 3. Evidence of clinically significant cardiovascular, neurological, psychiatric, pulmonary, renal, hepatic, endocrine, metabolic, hematological disorders, gastrointestinal disease or immunodeficiency that would compromise safety, interpretation of results, or protocol adherence. 4. Documented history of anaphylaxis or evidence of helminthic parasitic infection as evidenced by stools being positive for a pathogenic organism according to local guidelines. 5. Pregnant or nursing (lactating) women.

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### 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Remibrutinib 25 mg twice a day (b.i.d.); Omalizumab (Trade name: Xolair; ATC code:

R03DX05) 300 mg every 4 weeks (q4w) as the active control. **Route of Administration:** Oral for remibrutinib (and its matching placebo tablet); Subcutaneous (s.c.) for omalizumab (and its matching placebo injection). **Duration of Treatment:** 52 weeks double-blind treatment period followed by an optional 52-week open-label extension to assess long-term efficacy, safety and tolerability of remibrutinib 25 mg b.i.d.

**5.2 Concomitant & Prohibited Therapy Permitted:** Stable, local label-approved standard dose of a second-generation H1-AH ("background therapy") must be maintained. Different second-generation H1-AHs are allowed on an as-needed basis as "rescue therapy" for unbearable CSU flare-ups. **Prohibited:** Use of prior or concurrent biologics including anti-IgE therapies.

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days) | Key Activities/Procedures                                                                                                                                                                                                                                                  |
|--------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Screening    | Up to 4 weeks | Informed Consent, determination of baseline disease activity, wash-out of prohibited medications.                                                                                                                                                                          |
| Baseline     | Day 1         | Randomization (1 of 4 treatment arms), Double-dummy treatment initiation.                                                                                                                                                                                                  |
| Interim      | Weeks 12, 24  | Efficacy Assessments (UAS7, ISS7, HSS7), safety labs.                                                                                                                                                                                                                      |
| End of Study | Week 52       | Final Core Phase assessments. Eligible participants may transition to an optional 52-week open-label extension phase. Participants not entering the extension will have a 16-week safety follow-up (participants completing the extension have a 4-week safety follow-up). |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Absolute change from baseline in Weekly Urticaria Activity Score (UAS7), where UAS7 (range 0-42) is the sum of the Weekly Hives Severity Score (HSS7) and the Weekly Itch Severity Score (ISS7) at Week 12. **Measurement Method:** Clinical assessment via the Urticaria Patient Daily Diary (UPDD).

**7.2 Secondary & Exploratory Endpoints** 1. Achievement of UAS7 = 0 (yes/no) representing complete remission. 2. Improvement of severity of itch, assessed as absolute change from baseline in ISS7 score. 3. Improvement of severity of hives, assessed as absolute change from baseline in HSS7 score. 4. Safety and tolerability comparisons between remibrutinib, omalizumab, and placebo over 24 and 52 weeks.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring for AEs, including AESIs (Adverse Events of Special Interest) such as allergic reactions/anaphylaxis, injection site reactions, and

parasitic (helminth) infections. **Serious Adverse Events (SAEs):** Routine recording and monitoring per Good Clinical Practice (GCP) guidelines. **Data Monitoring Committee (DMC):** Internal and regulatory oversight provided by Novartis safety committees and national ethical boards.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy analyses; Safety Set for AE/SAE frequency reporting. **Sample Size Justification:** A target enrollment of 471 participants (approximately \$N=468\$ to be randomized) is appropriately powered to detect significant differences between remibrutinib, placebo, and omalizumab based on historical variances in UAS7, ISS7, and HSS7. **Handling of Missing Data:** Standard imputation methodology utilized per protocol (e.g., managing missing UPDD entries).

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with the International Council for Harmonisation (ICH) E6(R2) Good Clinical Practice (GCP) and the Declaration of Helsinki.

**Monitoring Plan:** Regular onsite and remote monitoring for data integrity, focusing on eCRF locking and compliance with double-dummy blinding protocols. **Audit Trail:** Stringent logging of Patient Daily Diary data and compliance with washout requirements.

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## 11. Study Identifiers & Governance

**Primary ID:** CLOU064A2304 **Registry Number:** NCT06042478 **Posting ID:** 139866335997781069 **Sponsor/Lead Organization:** Novartis **Study Title:** Phase 3b Study to Assess the Efficacy, Safety, and Tolerability of Remibrutinib in Comparison to Placebo, With Omalizumab as Active Control, in Adult CSU Patients, Followed by an Open-label 52-week Optional Extension. **Official Regulatory Title:** A Global, Multicenter, Randomized, Double-blind, Double-dummy, Parallel-group, Phase 3b Study to Assess the Efficacy, Safety, and Tolerability of Remibrutinib 25 mg b.i.d. in Comparison to Placebo With Omalizumab 300 mg Every 4 Weeks as Active Control Over 52 Weeks in Adult Patients With Chronic Spontaneous Urticaria Inadequately Controlled by Second Generation H1-antihistamines and an Open-label 52-week Optional Extension to Assess Long-term Efficacy, Safety and Tolerability of Remibrutinib 25 mg b.i.d. **Trial Status:** Active but not currently recruiting (ACTIVE\_NOT\_RECRUITING)

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## 12. Clinical Framework (CSU Specific)

**Primary Condition / Disease Area:** Chronic Spontaneous Urticaria (CSU) **Diagnosis Threshold:** Inadequately controlled by H1-antihistamines (UAS7  $\geq 16$ , ISS7  $\geq 6$ , HSS7  $\geq 6$ ) **Medical Methodology:** BTK inhibitor therapy (Remibrutinib) vs anti-IgE biologic therapy (Omalizumab) **Optimization Target:** Significant reduction in hives and itch (Targeting complete remission UAS7  $= 0$ )

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## 13. Study Procedures & Methodology

**Management Pathway:** Double-blind, double-dummy controlled evaluation period transitioning to an optional open-label extension where patients receive open-label remibrutinib only (no background therapy). **Education Protocol (HCP):** Site initiation and training regarding the double-dummy administration (oral and s.c. injections) **Education Protocol (Patient):** Comprehensive instructions on self-reporting via the Urticaria Patient Daily Diary (UPDD) and protocol regarding rescue vs. background medications **Quality Audit Mechanism:** Assessment of diary compliance; maximum 1 missing entry allowed in the 7 days prior to randomization

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Up to 112 weeks total study duration (4 weeks screening, 52 weeks core treatment, 52 weeks extension, and safety follow-up). **Follow-up details:** Participants not entering the extension have a 16-week treatment-free safety follow-up. Those entering the extension have a 4-week treatment-free safety follow-up at the end. **Onsite Visit Cadence:** Randomization, Baseline, and specific interval assessments (Week 12, 24, 52). **Remote Monitoring Frequency:** Intermittent monitoring based on site and local trial capabilities. **Checklist Review Interval:** Continuous daily tracking via the UPDD.

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|------|-----|-----|
| Role                     | Name   | Signature | Date          | :-- | :--  | :-- | :-- |
| Principal Investigator   | [Name] | ____      | [DD-MMM-YYYY] |     |      |     |     |
| Clinical Operations Lead | [Name] | ____      | [DD-MMM-YYYY] |     | Lead |     |     |
| Statistician             | [Name] | ____      | [DD-MMM-YYYY] |     |      |     |     |